

Autoimmune Pancreatitis (MONDO:0015175): A Comprehensive Disease Characterization

Summary

Autoimmune pancreatitis (AIP) is a rare, corticosteroid-responsive fibroinflammatory form of chronic pancreatitis that is now understood to comprise **three immunologically distinct subtypes**. **Type 1 AIP** — lymphoplasmacytic sclerosing pancreatitis (LPSP) — is the pancreatic manifestation of the systemic **IgG4-related disease (IgG4-RD)** and accounts for the large majority of cases (~94% of surgically confirmed AIP). It is defined by elevated serum IgG4, storiform fibrosis, obliterative phlebitis, dense IgG4+ plasma-cell infiltration, and frequent multi-organ (biliary, salivary/lacrimal, renal, retroperitoneal, aortic) involvement. **Type 2 AIP** — idiopathic duct-centric pancreatitis (IDCP) — is IgG4-negative, pancreas-restricted, histologically defined by the neutrophilic granulocytic epithelial lesion (GEL), and associated in ~one-third of cases with inflammatory bowel disease, especially ulcerative colitis. **Type 3 AIP** is a newly recognized, drug-induced entity triggered by immune-checkpoint-inhibitor (ICI) cancer immunotherapy.

Mechanistically, Type 1 AIP is driven by a self-sustaining loop between **oligoclonally expanded CD4+SLAMF7+ cytotoxic T lymphocytes (CTLs)** — which possess both direct cytotoxic and profibrotic properties — and **B-lineage cells/plasmablasts** that present antigen and differentiate into IgG4-secreting plasma cells. An upstream **innate-immunity axis** (neutrophil extracellular traps → plasmacytoid dendritic cells → IFN-α) drives IgG4-class switching and is necessary for disease in the principal MRL/Mp poly(I:C) mouse model. Type 2 AIP is instead neutrophil-centric, and Type 3 (ICI-related) is characterized by acinar-centric T-cell infiltration evolving from CD4+ to CD8+ predominance. Genetic susceptibility is **polygenic/multifactorial** (HLA-DRB1, FCGR2B, CTLA4, FCRL3) rather than Mendelian, and etiology combines genetic predisposition, environmental/occupational triggers (industrial fumes, asbestos), lifestyle factors (smoking), and — for Type 3 — drug exposure.

Clinically, AIP typically presents in **older men** (male:female $\approx$ 3:1; mean age $\sim$ 65–68 in Japanese surveys) with **painless obstructive jaundice and weight loss** that mimic pancreatic cancer, making the distinction from malignancy the central diagnostic challenge. Diagnosis follows the **ICDC** or **HISORt** frameworks integrating histology, imaging, serology, other-organ involvement, and steroid response. Prognosis is favorable ($\sim$ 85% 10-year survival) but **Type 1 relapses in $\sim$ 40%** (rising to $\sim$ 57% cumulatively at 10 years), whereas Type 2 rarely relapses. First-line therapy is corticosteroids; rituximab and immunomodulators manage refractory/relapsing disease, and the anti-CD19 antibody **inebilizumab became the first FDA-approved therapy for IgG4-RD in 2025** (MITIGATE phase 3). This report synthesizes 20 confirmed findings from 52 reviewed papers into a comprehensive knowledge-base entry.

Section 1 — Disease Information

Overview. AIP is a rare, distinct form of chronic pancreatitis characterized by pancreatic inflammation with a fibroinflammatory basis and, in most cases, an autoimmune/immune-mediated pathogenesis and dramatic response to corticosteroids. It represents approximately **2% of chronic pancreatitis cases** (PMID: 25099388). The disease is now formally divided into three subtypes (PMID: 40364113):

- **AIP-1 / Type 1** — lymphoplasmacytic sclerosing pancreatitis (LPSP); pancreatic manifestation of IgG4-RD
- **AIP-2 / Type 2** — idiopathic duct-centric pancreatitis (IDCP)
- **AIP-3 / Type 3** — immune-checkpoint-inhibitor (ICI)-induced

Key identifiers. - **Mondo:** MONDO:0015175 - **MeSH:** Autoimmune Pancreatitis (D000081012) - **ICD-11:** DC31 (chronic pancreatitis grouping); commonly coded under K86.1 (ICD-10, other chronic pancreatitis) / K85–K86 pancreatic disease - **Orphanet:** Autoimmune pancreatitis (classified within IgG4-related disease / AIP) - **OMIM:** No single Mendelian OMIM entry — AIP is a complex/polygenic disease, not a monogenic disorder

Synonyms and alternative names. Lymphoplasmacytic sclerosing pancreatitis (LPSP; Type 1); idiopathic duct-centric pancreatitis (IDCP; Type 2); IgG4-related pancreatitis; sclerosing pancreatitis; non-alcoholic duct-destructive chronic pancreatitis.

Source of information. The information in this report is derived predominantly from **aggregated disease-level resources** — nationwide epidemiological surveys (Japan), multicenter cohorts, systematic reviews/meta-analyses, GWAS, mechanistic immunology studies, and a phase 3 randomized trial — supplemented by individual case reports. It is not primarily EHR-derived.

Section 2 — Etiology

Disease causal factors. AIP etiology is **multifactorial**, involving a complex interplay of **genetic predisposition, environmental triggers, and dysregulated adaptive immunity** (PMID: 41728618). No single Mendelian cause exists. Type 3 is uniquely **drug-induced** (ICIs) (PMID: 42467391).

"The pathogenesis of IgG4-SC involves a complex interplay of genetic predisposition, environmental triggers (e.g., industrial vapors, dust, gases, fumes, and asbestos), and dysregulated adaptive immunity." (PMID: 41728618)

Genetic risk factors. A Japanese GWAS of 835 IgG4-RD patients vs 1,789 controls identified two genome-wide-significant susceptibility loci: **HLA-DRB1** and **FCGR2B** (PMID: 38229354). Earlier associations include the **HLA DRB1*0405–DQB1*0401** haplotype and **FCRL3** (Fc-receptor-like 3) polymorphisms (PMID: 18341485). **CTLA4** +6230 G/G increased AIP risk (OR 2.48, p=0.011), and the +49A/A and +6230A/A genotypes increased relapse risk (OR 5.45 and 12.66) with elevated serum soluble CTLA4 (8.9 vs 2.9 ng/mL, p<0.001) (PMID: 18341485). Targeted high-throughput sequencing implicated **P2RX3** and **TOP1** (surviving Bonferroni correction) among 11 candidate genes for Type 1 AIP susceptibility (PMID: 28955865).

Environmental risk factors. Occupational/industrial exposures — **industrial vapors, dust, gases, fumes, and asbestos** — are implicated triggers (PMID: 41728618). In an AIP cohort, **blue-collar profession and smoking** were potential risk factors for adverse metabolic outcomes (PMID: 35807009). **Older age** and **male sex** are strong demographic risk factors (see Section 9).

Protective factors. No well-established genetic protective variants have been defined for AIP. Environmental protective factors are not established. Notably, **serum IgG4 normal at diagnosis or normalized after steroids is protective against relapse** ([PMID: 42185503](#)), and **corticosteroid maintenance** reduces relapse.

Gene–environment interactions. The proposed model is that environmental/innate triggers (e.g., NET-inducing stimuli, industrial exposures) act on a genetically primed immune background (HLA-DRB1, FCGR2B, CTLA4, FCRL3) to unleash the pathogenic CD4+ CTL/plasmablast axis. Molecular mimicry (e.g., *Helicobacter pylori* plasminogen-binding protein homology with human carbonic anhydrase) has been **hypothesized but not proven**.

Section 3 — Phenotypes

Cardinal presenting phenotypes. The most common presentation overall is **painless obstructive jaundice with weight loss**, closely mimicking pancreatic cancer ([PMID: 38516247](#), [PMID: 32234378](#)).

"The clinical manifestations of AIP mainly include painless jaundice and weight loss." ([PMID: 38516247](#))

In the 2016 Japan survey, **63% of patients were symptomatic**, roughly half of whom had jaundice ([PMID: 31872350](#)).

Phenotype	Type	HPO suggestion	Frequency / Notes
Obstructive jaundice	Clinical sign	HP:0000952 (Jaundice)	~50% of symptomatic; independent risk factor for diabetes (PMID: 35807009)
Weight loss	Symptom	HP:0001824	Common presenting feature (PMID: 38516247)
Abdominal pain	Symptom	HP:0002027	Variable; usually mild (contrast with acute pancreatitis)
Pancreatic exocrine insufficiency (PEI)	Lab/functional	HP:0001738	72.7% at diagnosis, 63.5% at follow-up (PMID: 35807009)
Diabetes mellitus (endocrine insufficiency)	Lab abnormality	HP:0000819	Prevalence 32.8% at diagnosis; cumulative incidence 17.9% (PMID: 35807009)
Elevated serum IgG4	Lab abnormality	HP:0030355 (Increased circulating IgG level)	86–88% in Type 1 (Japan surveys)
Diffuse pancreatic enlargement ("sausage" pancreas)	Imaging sign	HP:0012093-related	Type 1 typical imaging
Sialadenitis / lacrimal gland swelling	Physical manifestation	HP:0000163 / HP:0000509-related	IgG4-RD systemic feature (Type 1)
Eosinophilia	Lab abnormality	HP:0001880	Reported in subset of IgG4-RD (PMID: 38407323)

Phenotype characteristics. Onset is **adult/geriatric** (see Section 8). Progression is typically **insidious/chronic** with an **episodic/relapsing** course in Type 1. Severity is variable; most cases respond dramatically to steroids.

"PEI prevalence at diagnosis was 72.7% and was 63.5% at follow-up. The cumulative incidence of DM was 17.9%, with a prevalence of DM at diagnosis of 32.8%." (PMID: 35807009)

Quality-of-life impact. Formal EQ-5D/SF-36/PROMIS data specific to AIP were **not identified**. Functional impact is driven largely by exocrine insufficiency (malabsorption, weight loss), new-onset diabetes, and, in relapsing disease, repeated hospitalizations and cumulative organ damage.

Section 4 — Genetic / Molecular Information

Causal genes. AIP is **not monogenic**; there are no causal Mendelian genes. Instead, **susceptibility loci** confer polygenic risk:

Gene / locus	HGNC	Evidence	Effect
HLA-DRB1 (esp. DRB1*0405–DQB1*0401 haplotype)	HGNC:4948	GWAS genome-wide significant (PMID: 38229354); haplotype association (PMID: 18341485)	Antigen presentation susceptibility
FCGR2B	HGNC:3618	GWAS genome-wide significant (PMID: 38229354)	Inhibitory Fc receptor; B-cell regulation
CTLA4	HGNC:2505	+6230 G/G OR 2.48 for risk; relapse OR up to 12.66 (PMID: 18341485)	T-cell checkpoint
FCRL3	HGNC:18506	Polymorphism association (PMID: 18341485)	B-cell regulation
P2RX3, TOP1	HGNC:8535 / HGNC:11986	Candidate (Bonferroni-significant) (PMID: 28955865)	Susceptibility markers

"Two susceptibility loci for IgG4-related disease were identified. Both FCGR2B and HLA loci might have important roles in IgG4-related disease development." (PMID: 38229354)

Pathogenic variants. Because AIP is polygenic, ACMG/AMP pathogenicity classification does not apply in the Mendelian sense. Associated variants are **common susceptibility polymorphisms/haplotypes** (e.g., HLA class II alleles, CTLA4 SNPs rs231775 [+49A/G], +6230 A/G) rather than rare pathogenic mutations. These are **germline** in origin. Functional consequence is **immune dysregulation** (altered antigen presentation, impaired T-cell checkpoint/inhibitory-receptor signaling) rather than classic loss/gain of function in a structural protein.

Autoantigens (molecular targets). Four candidate autoantigens have been described in IgG4-RD: **annexin A11, galectin-3 (LGALS3), laminin 511-E8, and prohibitin 1** (PMID: 38332916). Anti-galectin-3 autoantibodies were detected in 13.5% of IgG4-related cholangitis patients but not in primary sclerosing cholangitis controls.

"Four autoantigens have recently been described in IgG4-RD: annexin A11, galectin-3, laminin 511-E8, and prohibitin 1." (PMID: 38332916)

Modifier genes. CTLA4 genotypes modify **relapse risk** (+49A/A OR 5.45; +6230A/A OR 12.66) (PMID: 18341485). Candidate relapse-modifying variants (HLA-C, CXCR3, CACNA1C) were reported by targeted sequencing (PMID: 28955865).

Epigenetic information & chromosomal abnormalities. No established disease-specific DNA methylation, histone-modification, or chromosomal-abnormality signatures for AIP were identified. This is a **knowledge gap**.

Section 5 — Environmental Information

Environmental factors. Industrial/occupational exposures — **industrial vapors, dust, gases, fumes, and asbestos** — are implicated as triggers of IgG4-RD/AIP (PMID: 41728618). **Blue-collar occupation** was associated with adverse metabolic outcomes (PMID: 35807009).

Lifestyle factors. Smoking is the best-supported lifestyle risk factor, identified as a potential contributor to diabetes/exocrine-insufficiency outcomes in AIP (PMID: 35807009) and a recognized risk factor for chronic pancreatitis more broadly.

Infectious agents. No single infectious agent has been established as causal. Molecular mimicry involving *Helicobacter pylori* plasminogen-binding protein (homology with human carbonic anhydrase II / ubiquitin-protein ligase) has been hypothesized but **not proven**. AIP is not a transmissible/zoonotic disease.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain (Type 1 AIP / IgG4-RD)

1. Genetic priming (HLA-DRB1, FCGR2B, CTLA4, FCRL3 risk alleles)
+ environmental/innate trigger
→ leads to →
2. Innate activation: tissue stimulus induces NEUTROPHIL EXTRACELLULAR TRAPS (NETs)
→ NETs stimulate →
3. PLASMACYTOID DENDRITIC CELLS (pDCs) accumulate in pancreas and produce IFN- α
[NECESSARY step: pDC depletion / IFN- α blockade prevents AIP in mouse model]
→ IFN- α (with BAFF) drives →
4. B-CELL activation & IgG4-CLASS SWITCHING → oligoclonal expansion of IgG4+ PLASMA BLASTS/plasma cells
→ plasmablasts present antigen to →
5. CD4+SLAMF7+ CYTOTOXIC T LYMPHOCYTES (CTLs) – oligoclonally expanded – infiltrate lesions
[self-sustaining loop: CTLs ↔ plasmablasts via continuous antigen presentation]
→ CD4+ CTLs exert DUAL effects →
 (a) direct cytotoxicity → acinar/epithelial cell death (TISSUE DAMAGE)
 (b) secretion of profibrotic factors (e.g., TGF- β) → activate fibroblasts
→ results in →
6. STORIFORM FIBROSIS + OBLITERATIVE PHLEBITIS + dense lymphoplasmacytic infiltrate
→ produces →
7. Mass-forming pancreatic enlargement, ductal narrowing → OBSTRUCTIVE JAUNDICE,
exocrine & endocrine insufficiency (CLINICAL MANIFESTATION)

Branch — Type 2 AIP (IDCP): IgG4-independent; **neutrophil-mediated granulocytic epithelial lesions (GEL)** destroy duct epithelium → pancreas-restricted duct-centric inflammation (frequently co-occurring with IBD).

Branch — Type 3 AIP (ICI-related): ICI blockade of PD-1/PD-L1/CTLA-4 removes T-cell checkpoint restraint → **acinar-centric T-cell infiltration evolving from CD4+ to CD8+ predominance** → often asymptomatic hyperlipasemia, with pancreatic atrophy/diabetes as sequelae (PMID: 41922528).

Detail by category

Immune system involvement — the core mechanism. The traditional Th2 (IL-4/IL-10) paradigm is now considered insufficient to explain tissue destruction and fibrosis. Instead, **oligoclonally expanded CD4+ cytotoxic T lymphocytes (CTLs)** bearing **SLAMF7** are the central effectors, with dual cytotoxic and profibrotic properties (PMID: 41766862):

"Emerging data highlight the extensive, oligoclonally expanded infiltration of CD4+ cytotoxic T lymphocytes (CTLs) deep within lesions. These cells possess dual cytotoxic and profibrotic properties." (PMID: 41766862)

These CTLs are sustained by **continuous antigen presentation by B-lineage cells, particularly plasmablasts** (PMID: 27667138), explaining why B-cell depletion (rituximab, inebilizumab) is effective. The CD4+SLAMF7+ population expands in patients and **decreases after glucocorticoid treatment** (PMID: 29499100). A distinctive CD4+ response dominated by **Th2, T-follicular-helper (Tfh), and regulatory T cells (Tregs)** drives B-cell activation and IgG4+ plasmablast expansion (PMID: 41728618). M2 macrophages also contribute (PMID: 35737955).

"A distinctive CD4+ T-cell response, dominated by T-helper 2 (Th2), follicular helper T (Tfh) cells, and regulatory T cells (Tregs), drives B-cell activation, oligoclonal expansion of IgG4+ plasmablasts, and progressive fibrosis." (PMID: 41728618)

Innate immunity (upstream trigger). In the MRL/Mp mouse, disease develops in parallel with pancreatic accumulation of **IFN-α-producing plasmacytoid dendritic cells (pDCs)**; pDC depletion and IFN-α blockade **prevent AIP**, proving necessity (PMID: 26297761). **Neutrophil extracellular traps (NETs)** stimulate pDCs to make IFN-α; human patient pDCs cultured with NETs produced greatly increased IFN-α and induced B cells to make **IgG4 (but not IgG1)** (PMID: 26297761). **SLAMF7 on CD8+ T cells** is also increased in AIP (PMID: 37661465).

"patient pDCs cultured in the presence of NETs produced greatly increased levels of IFN- α and induced control B cells to produce IgG4 (but not IgG1)" (PMID: 26297761)

Cellular processes. Chronic inflammation, epithelial/acinar apoptosis and cytotoxic killing, fibroblast activation, and progressive fibrosis.

Tissue-damage mechanisms. Fibrosis (storiform pattern), obliterative phlebitis, lymphoplasmacytic destruction. In Type 2, neutrophilic duct destruction (GEL). Reduced ductal CFTR expression contributes to acinar dysfunction; CFTR correctors (C18) rescued function and reduced inflammation in mice (PMID: 28634110).

Molecular profiling. Circulating **plasmablast expansion** is a robust cellular biomarker (see Section 10). Cytokine signatures (IFN- γ , IL-10, IL-13, IL-5, IL-21) correlate with disease activity (PMID: 42277154).

Suggested ontology terms. GO:0002250 (adaptive immune response), GO:0001909 (leukocyte-mediated cytotoxicity), GO:0030198 (extracellular matrix organization / fibrosis), GO:0006954 (inflammatory response), GO:0032606 (type I interferon production). Cell types: CL:0000625 (CD8-positive cytotoxic T cell), CL:0000624 (CD4-positive T cell), CL:0000980 (plasmablast), CL:0000784 (plasmacytoid dendritic cell), CL:0000775 (neutrophil), CL:0000890 (M2 macrophage).

Section 7 — Anatomical Structures Affected

Primary organ. The **pancreas** (UBERON:0001264) is the primary affected organ in all subtypes.

Secondary/systemic involvement (Type 1 / IgG4-RD). IgG4-RD is a systemic mass-forming fibroinflammatory condition affecting nearly every organ (PMID: 26672716):

"The organs most frequently involved are the pancreas (autoimmune pancreatitis (AIP), salivary and lacrimal glands (Mickulicz disease and sclerosing sialadenitis), biliary tree (sclerosing cholangitis or cholecystitis), retroperitoneum (retroperitoneal fibrosis), aorta (periaortic fibrosis), kidneys (interstitial nephritis) and thyroid (Riedel thyroiditis)." (PMID: 26672716)

Structure	UBERON	Manifestation
Pancreas	UBERON:0001264	AIP (primary)
Bile ducts / biliary tree	UBERON:0002394	IgG4-related sclerosing cholangitis
Salivary glands	UBERON:0001044	Sclerosing sialadenitis (Mikulicz)
Lacrimal glands	UBERON:0001817	Dacryoadenitis
Kidney	UBERON:0002113	Tubulointerstitial nephritis
Retroperitoneum	UBERON:0003693	Retroperitoneal fibrosis
Aorta	UBERON:0000947	Periaortitis
Thyroid	UBERON:0002046	Riedel thyroiditis

Extrapancreatic lesions were detected in **60% of AIP patients** in the 2016 Japan survey ([PMID: 31872350](#)). **Type 2 AIP is pancreas-restricted** (except rare IBD/sialadenitis) ([PMID: 34670874](#), [PMID: 32825945](#)).

Tissue/cell level. Pancreatic **ductal and acinar epithelium** (targets), with infiltration by plasma cells, plasmablasts, CD4+/CD8+ T cells, pDCs, neutrophils (Type 2), and M2 macrophages. **Subcellular:** ER (secretory pathway of plasma cells) and ductal apical membrane (CFTR). **Body system:** digestive/endocrine.

Localization / lateralization. The pancreas can be diffusely enlarged ("sausage" pancreas, Type 1) or focally involved (mimicking cancer). Not a lateralized organ; involvement is described as diffuse vs focal/segmental.

Section 8 — Temporal Development

Onset. Adult to geriatric. Mean age at diagnosis ~64.8 years (2016 Japan survey; mean age 68.1) ([PMID: 31872350](#)); ~59–66 in other cohorts. Onset pattern is **insidious/chronic/subacute**, frequently presenting as painless jaundice. Type 2 tends to occur in somewhat younger patients than Type 1.

Progression. Type 1 follows a **relapsing-remitting** course. Cumulative relapse rates in a 10-year Type 1 cohort were **11.0% / 26.9% / 38.3% / 50.0% / 56.9% at 1/3/5/7/10 years** ([PMID: 42185503](#)):

"Cumulative relapse rates at 1, 3, 5, 7, and 10 years were 11.0%, 26.9%, 38.3%, 50.0%, and 56.9%." (PMID: 42185503)

Progression is generally **slow**; end-stage disease features pancreatic atrophy, exocrine insufficiency, and diabetes.

Patterns. Remission is typically **treatment-induced** (corticosteroids). Type 2 AIP relapse is uncommon (favorable). Corticosteroid **discontinuation independently predicts relapse**, while normal/normalized serum IgG4 is protective (PMID: 42185503). Critical intervention window: early steroid therapy induces remission and prevents irreversible fibrotic damage.

Section 9 — Inheritance and Population

Epidemiology (Japan nationwide surveys — the best-characterized data):

Year	Prevalence / 100,000	Incidence / 100,000/yr	M:F	Mean age	High serum IgG4
2007 (PMID: 22466167)	2.2	0.9	3.7	63.0	87.6%
2011 (PMID: 25815647)	4.6	1.4	3.2	66.3	86.4%
2016 (PMID: 31872350)	10.1	3.1	2.94	68.1	—

"The estimated number of AIP patients in 2016 was 13,436, with an overall prevalence rate of 10.1 per 100,000 persons. The estimated number of newly diagnosed patients was 3984, with an annual incidence rate of 3.1 per 100,000 persons." (PMID: 31872350)

Prevalence **more than doubled from 2011 to 2016**, reflecting rising recognition. AIP accounts for ~2% of chronic pancreatitis (PMID: 25099388).

Inheritance pattern. Multifactorial / polygenic — NOT Mendelian. No AD/AR/X-linked/mitochondrial inheritance. Penetrance, expressivity, anticipation, mosaicism, founder effects, consanguinity, and carrier frequency are **not applicable** in the classical genetic sense.

Demographics. Strong male predominance (M:F ~3:1), **elderly onset** (Type 1). Subtype distribution among surgically confirmed AIP: **Type 1 ~94%, Type 2 ~6%** (PMID: 39169289) — though Type 2 is relatively more common in Western/younger/IBD populations. Type 1 is more prevalent in East Asia; Type 2 is proportionally more common in Europe/North America.

"The male-to-female sex ratio was 2.94, the mean age was 68.1, and mean age at diagnosis was 64.8." (PMID: 31872350)

Section 10 — Diagnostics

Diagnostic frameworks. Diagnosis is established by the **International Consensus Diagnostic Criteria (ICDC)** or **HISORt** (Histology, Imaging, Serology, Other organ involvement, Response to therapy) (PMID: 25099388), and the Japan Pancreas Society (JPS) criteria (PMID: 40996454).

"Diagnosis of AIP is established according to the international consensus diagnostic criteria (ICDC) or HISORt (mnemonic standing for histology, imaging, serology, other organ involvement and response to therapy) criteria." (PMID: 25099388)

In **Type 1**, typical imaging changes can suffice even with negative histology; in **Type 2**, histologic evidence (GEL) is required (PMID: 37947862).

"In type 1 AIP, typical imaging changes are sufficient to establish the diagnosis even with negative histology, whereas for type 2 AIP, histologic evidence is required." (PMID: 37947862)

Serology / biomarkers. - **Serum IgG4:** >2× ULN suggestive, >4× ULN **highly specific**; **IgG4/IgG1 ratio >0.24** is discriminatory (PMID: 41728618). However, sensitivity is limited — ~13–16% of patients have normal serum IgG4, and in surgical series Type 1 sensitivity can be ~43% (PMID: 24817416, PMID: 39169289). - **Circulating plasmablasts:** a robust IgG4-independent

biomarker — markedly elevated (median 4,698/mL vs 592/mL untreated disease controls, 94/mL healthy; $p < 0.001$); **36% of IgG4-RD patients with NORMAL serum IgG4 still had elevated plasmablasts**; levels fall with rituximab and rise at flare (PMID: 24817416).

"The IgG4-RD patients had substantially elevated total plasmablast counts (median 4698/mL, range 610-79524/mL) compared to both untreated disease controls (median 592/mL, range 19-4294/mL; $p < 0.001$) and healthy controls (median 94/mL, range 1-653/mL; $p < 0.001$)." (PMID: 24817416)

- **Type 2 has no established serum marker** — diagnosis requires histologic GEL (PMID: 34670874).

"Since there are currently no established serum markers, the diagnosis of type 2 AIP is highly challenging and requires the tissue confirmation of neutrophilic injury to the pancreatic ducts, a finding designated as a granulocytic epithelial lesion." (PMID: 34670874)

Imaging. Diffuse pancreatic enlargement ("sausage" pancreas) with delayed/rim enhancement and narrowed main pancreatic duct (Type 1). **18F-FDG PET/CT** aids diagnosis, staging of systemic involvement, and treatment monitoring (PMID: 26672716). **Endoscopic ultrasound (EUS)** with EUS-guided core biopsy is the main modality for tissue sampling and for differentiating AIP from cancer (PMID: 40996454).

Histopathology. Type 1: storiform fibrosis, obliterative phlebitis, dense lymphoplasmacytic infiltrate rich in IgG4+ plasma cells (PMID: 28747608). Type 2: granulocytic epithelial lesion (GEL) — neutrophilic duct epithelial injury (PMID: 34670874).

Differential diagnosis. The critical differential is **pancreatic ductal adenocarcinoma**; also cholangiocarcinoma, primary sclerosing cholangitis, and other chronic pancreatitis (PMID: 40996454, PMID: 40191403).

Genetic testing. Not part of routine diagnosis (polygenic disease). HLA/CTLA4 genotyping is research-only.

Section 11 — Outcome / Prognosis

Survival. Favorable. **10-year overall survival 85.5%** in a corticosteroid-treated Type 1 cohort; corticosteroid maintenance associated with better survival (PMID: 42185503).

Relapse (key prognostic issue). Type 1 relapses in ~40–42% (PMID: 40773035, PMID: 41223493); cumulative ~57% at 10 years (PMID: 42185503). **Type 2 relapse is uncommon** (PMID: 37826920).

"AIP primarily consists of type 1 and type 2, with relapse being a significant problem mainly associated with type 1 AIP, which has a high relapse rate of approximately 40%, whereas type 2 AIP has significantly lower relapse rates." (PMID: 40773035)

Prognostic factors for relapse: younger age (<60 y; OR 1.7, 95% CI 1.1–2.7, p=0.022) (PMID: 41223493); persistent/insufficiently declining serum IgG4, diffuse pancreatic swelling, proximal biliary/renal involvement (PMID: 40773035); corticosteroid discontinuation (PMID: 42185503).

"The overall relapse rate was 41.9 %, being higher in the <60 years group (46.2 %) than in the ≥60 years group (37.0 %). Age <60 years was significantly associated with increased relapse risk (OR = 1.7; 95 % CI: 1.1-2.7, P = 0.022)." (PMID: 41223493)

Morbidity/complications. Exocrine insufficiency (up to 72.7%), diabetes (~33% prevalence), biliary strictures, and progressive fibrosis. HbA1c worsened and serum albumin declined (mean −0.27 g/dL) by 10 years (PMID: 42185503).

Malignancy risk. Overall malignancy risk **not increased** (SIR 1.00, 95% CI 0.59–1.42), but **pancreatic cancer SIR numerically higher (1.98 overall; 2.96 at ≥5 years)** (PMID: 42185503). A nationwide Swedish study found autoimmune diseases collectively raise pancreatic-cancer risk (SIR 1.24 men, 1.19 women) (PMID: 41795138).

"Overall malignancy risk was not increased (SIR 1.00; 95%CI 0.59-1.42), while pancreatic cancer (PC) showed a numerically higher SIR (1.98 overall; 2.96 ≥ 5 years postdiagnosis)." (PMID: 42185503)

Section 12 — Treatment

First-line: corticosteroids. Oral corticosteroids are the standard first-line therapy, effective in the majority (NCIT: Corticosteroid Therapy) (PMID: 36293522, PMID: 38516247). Prolonged **maintenance glucocorticoid** reduces relapse (PMID: 40773035).

"The standard therapy for AIP is oral administration of corticosteroids. Rituximab (RTX) has also been proposed for induction of remission and maintenance therapy in relapsing AIP-1." (PMID: 36293522)

B-cell-targeted / immunomodulatory (relapsing/refractory disease). - **Rituximab** (anti-CD20; NCIT:C1702) — induction and maintenance in relapsing AIP-1 (PMID: 36293522). - Steroid-sparing immunomodulators: **azathioprine, methotrexate, mycophenolate mofetil** (off-label) (PMID: 40745228).

Inebilizumab (anti-CD19) — first FDA-approved therapy for IgG4-RD (2025). The phase 3 **MITIGATE** trial (n=135) showed inebilizumab significantly reduced recurrence (**10% vs 60% placebo**), annual exacerbation rate, and glucocorticoid need; serum IgG4 fell ~50% with persistent B-cell depletion over 52 weeks; more infections/lymphopenia but no treatment-associated deaths (PMID: 40745228).

"The treatment significantly reduced the risk of recurrence (10% vs. 60% under placebo), the annual exacerbation rate and the necessity for renewed administration of glucocorticoids." (PMID: 40745228)

"Following approval of inebilizumab by the U.S. Food and Drug Administration (FDA) for IgG4-RD in 2025" (PMID: 40745228)

Type 2 AIP. Responds well to glucocorticoids; **anti-TNF- α antibodies** are a promising alternative (PMID: 37826920).

"Patients with AIP-2 respond well to glucocorticoids, with anti-tumor necrosis factor-alpha antibodies as a promising alternative therapy." (PMID: 37826920)

Type 3 (ICI-related). Corticosteroid role controversial; management guidelines not established; ICI cessation and supportive care are used (PMID: 41922528).

Experimental targeted agents (under evaluation). Anti-SLAMF7 (elotuzumab), obexelimab (CD19), BTK inhibitors, JAK/STAT inhibitors, and T2-inflammation biologics ([PMID: 37858433](#)). Targeting the **plasmablast–B-cell lineage and CD4+SLAMF7+ CTL** axis is the most promising future direction ([PMID: 35737955](#)).

Supportive care. Pancreatic enzyme replacement (exocrine insufficiency), insulin/oral agents (diabetes), biliary stenting (obstructive jaundice).

Pharmacogenomics. CTLA4 genotypes modulate relapse risk ([PMID: 18341485](#)); no validated genotype-guided dosing protocol exists yet.

Section 13 — Prevention

Primary prevention. No established primary prevention (etiology is multifactorial/idiopathic). Modifiable risk-factor reduction (smoking cessation; avoiding industrial exposures) is reasonable but unproven. For **Type 3**, awareness of ICI risk allows monitoring but not avoidance without foregoing cancer therapy.

Secondary prevention (early detection). Vigilant differentiation of AIP from pancreatic cancer to avoid unnecessary resection ([PMID: 40191403](#)); serial serum IgG4 and imaging for early relapse detection.

Tertiary prevention (complication prevention). **Corticosteroid maintenance** to prevent relapse and irreversible fibrotic organ damage ([PMID: 40773035](#), [PMID: 42185503](#)); long-term surveillance with labs, imaging (MRCP/EUS), and endoscopy for biliary strictures ([PMID: 40191403](#)); monitoring for exocrine/endocrine insufficiency and pancreatic cancer.

Immunization, genetic screening, counseling. Not applicable (non-infectious, polygenic disease). No carrier/prenatal screening.

Section 14 — Other Species / Natural Disease

- **Taxonomy:** Human (NCBI:txid9606) is the disease context. Experimental disease is induced in **mouse** (NCBI:txid10090).
- **Naturally occurring animal disease:** No well-established naturally occurring analog of human AIP in companion animals or wildlife was identified in the reviewed literature.

(Lymphoplasmacytic/immune-mediated pancreatitis exists in dogs but is not established as an AIP ortholog.) — **Knowledge gap.**

- **Orthologous genes:** CTLA4, FCGR2B, and HLA (H2 in mouse) have murine orthologs used in models.
- **Zoonotic potential:** None — AIP is non-transmissible.

Section 15 — Model Organisms

Principal model: MRL/Mp mouse + poly(I:C). Autoimmune-prone MRL/Mp mice given repeated intraperitoneal **polyinosinic-polycytidylic acid [poly(I:C)]** develop experimental AIP with massive pancreatic architecture destruction, immune infiltration, fibrosis, multi-organ involvement, and elevated autoantibodies resembling human IgG4-RD ([PMID: 29512140](#)).

"autoimmune-prone MRL/Mp mice treated with repeated injection with polyinosinic-polycytidylic acid (poly (I:C)) provide an experimental model of AIP. These mice exhibit massive destruction of pancreatic architecture associated with pancreatic immune cell infiltration and fibrosis." ([PMID: 29512140](#))

Key limitation. *"Although mice lack the IgG4 Ab subtype"* ([PMID: 29512140](#)) — mice cannot recapitulate the defining IgG4 serology/plasma-cell feature, limiting translational fidelity for the IgG4 axis.

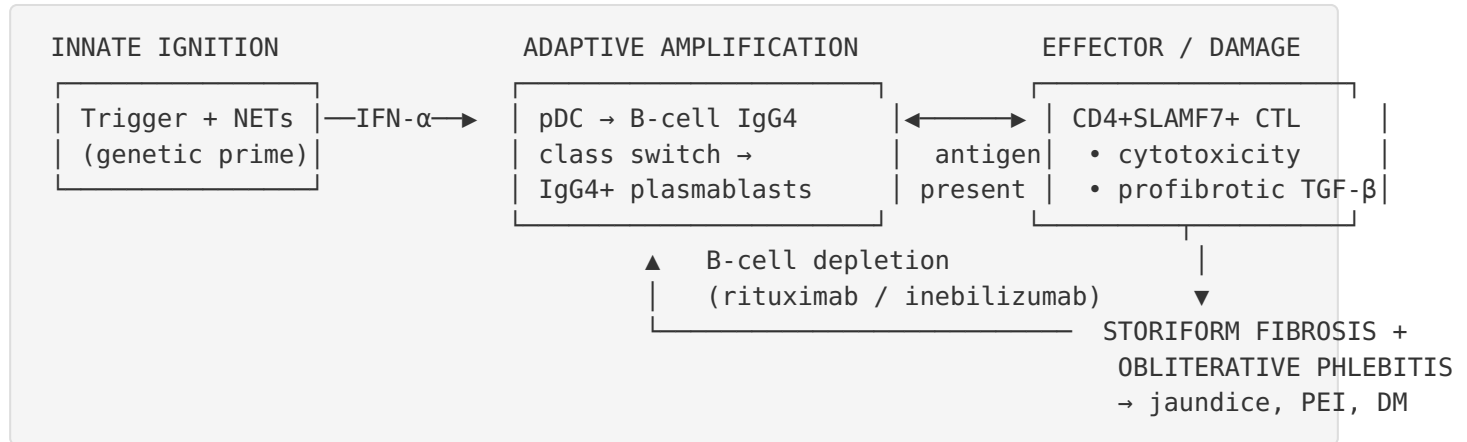
Other models. NOD/ShiLTJ mice and **BMP6-transduced** mice model Sjögren's/chronic pancreatitis; MRL/Mp models AIP with markedly reduced ductal **CFTR** expression — CFTR correctors (C18) rescued function and reduced inflammation ([PMID: 28634110](#)).

Applications. These models established the NET → pDC → IFN-α → IgG4 innate axis ([PMID: 26297761](#)), the role of SLAMF7+ CD8 T cells ([PMID: 37661465](#)), and the therapeutic potential of CFTR correction and steroid response.

Resources: MGI (mouse); model strains MRL/MpJ, NOD/ShiLTJ.

Mechanistic Model / Interpretation

The synthesized model positions Type 1 AIP as a **two-compartment autoimmune loop** ignited by innate immunity:



Two therapeutic insights follow directly from this architecture. First, because CD4+ CTLs are **sustained by plasmablast antigen presentation**, depleting the B-cell/plasmablast compartment collapses the entire loop — explaining why **rituximab** and now **inebilizumab** (anti-CD19, which reaches later B-lineage stages including plasmablasts) are effective. Second, the **innate IFN-α axis** offers an upstream target not yet exploited clinically. Type 2 and Type 3 are mechanistically distinct — neutrophil-driven and checkpoint-release T-cell-driven, respectively — and therefore require different management, underscoring why accurate subtyping is clinically essential.

Evidence Base

Domain	Key PMIDs	Contribution
Subtype definition & frequency	39169289 , 38516247 , 40364113	Established Type 1/2/3 taxonomy and 94%/6% split
Core immune mechanism	41766862 , 27667138 , 29499100 , 28747608	CD4+SLAMF7+ CTL / plasmablast loop; histopathology
Innate trigger	26297761 , 29512140	NET→pDC→IFN-α axis; mouse model
Genetics	38229354 , 18341485 , 28955865	HLA-DRB1, FCGR2B, CTLA4, FCRL3
Autoantigens	38332916	Annexin A11, galectin-3, laminin 511-E8, prohibitin 1
Epidemiology	31872350 , 25815647 , 22466167	Prevalence/incidence trends (Japan)
Diagnosis / biomarkers	25099388 , 37947862 , 24817416 , 41728618 , 40996454	ICDC/HISORT; IgG4 thresholds; plasmablasts
Prognosis / relapse	40773035 , 41223493 , 42185503	40% relapse; 85.5% 10-yr survival; PC risk
Treatment	36293522 , 40745228 , 37826920	Steroids, rituximab, inebilizumab (MITIGATE), anti-TNF
Type 2	34670874 , 37826920 , 32825945	GEL, IBD association, favorable prognosis
Type 3 (ICI)	42467391 , 41922528	Definition, incidence 0.5–5.7%, CD4→CD8 histology
Systemic / anatomy	26672716 , 32234378 , 40191403	Multi-organ IgG4-RD; biliary involvement
Clinical phenotype	35807009	Exocrine (72.7%)/endocrine (32.8%) insufficiency

The evidence base is internally consistent: mechanistic (mouse/in-vitro), genetic (GWAS), epidemiological (nationwide surveys), and therapeutic (phase 3 RCT) lines converge on a B-cell/plasmablast-CD4+ CTL model of Type 1 AIP, validated therapeutically by the success of B-cell depletion.

Limitations and Knowledge Gaps

1. **Epidemiology is Japan-centric.** The best prevalence/incidence data derive from Japanese nationwide surveys; Western/global incidence, and true Type 2 proportion outside Asia, are less precisely quantified.
2. **No monogenic cause / no ClinVar pathogenic variants.** AIP is polygenic; standard ACMG variant classification, penetrance, and carrier-frequency concepts do not apply. Epigenetic and chromosomal data are essentially absent.
3. **Autoantigen causality unproven.** The four candidate autoantigens (annexin A11, galectin-3, laminin 511-E8, prohibitin 1) are associations; a single dominant, disease-driving autoantigen has not been definitively established.
4. **Serum IgG4 imperfect.** ~13–16% of patients are seronegative; specificity is limited, and it correlates inconsistently with activity in fibrotic/localized subsets ([PMID: 42277154](#)).
5. **Mouse model lacks IgG4.** The MRL/Mp poly(I:C) model cannot recapitulate the defining IgG4 serology.
6. **Type 3 evidence is thin.** AIP-3 rests on small retrospective series with no consensus diagnostic/management guidelines.
7. **QoL data absent.** No AIP-specific EQ-5D/SF-36/PROMIS data were identified.
8. **Inebilizumab is new.** Long-term (>52-week) safety/efficacy and comparative data vs rituximab are not yet available.

Proposed Follow-up Experiments / Actions

1. **Validate circulating plasmablasts + IgG4/IgG1 ratio as a combined diagnostic/monitoring panel** in a prospective multi-ethnic cohort, particularly to capture IgG4-seronegative patients.
2. **Single-cell / spatial transcriptomics of AIP pancreatic tissue** to map the CD4+SLAMF7+ CTL-plasmablast niche and identify upstream antigen-presenting cell states (fills the "which autoantigen" gap).

3. **Test upstream innate blockade** (anti-IFN- α / anti-type-I-IFN receptor, or NET inhibitors) in the MRL/Mp model as a steroid-sparing strategy, given the demonstrated necessity of the pDC/IFN- α axis.
4. **Head-to-head or registry comparison of inebilizumab vs rituximab** for relapse prevention, with cost, infection risk, and durability endpoints.
5. **Establish international Type 2 and Type 3 registries** with standardized histologic/GEL and ICI-exposure criteria to define incidence, natural history, and management.
6. **Prospective pancreatic-cancer surveillance study** in Type 1 AIP (≥ 5 -year follow-up) to confirm/quantify the numerically elevated pancreatic-cancer SIR and define screening value.
7. **Epigenomic profiling** (methylation/ATAC-seq) of AIP tissue to fill the current absence of epigenetic data.

Consensus Answer

Autoimmune pancreatitis (AIP; MONDO:0015175) is a rare, corticosteroid-responsive fibroinflammatory chronic pancreatitis comprising three immunologically distinct subtypes — Type 1 (lymphoplasmacytic sclerosing pancreatitis, the pancreatic manifestation of IgG4-related disease, driven by a self-sustaining CD4+SLAMF7+ cytotoxic-T-lymphocyte/plasmablast loop with an upstream NET→plasmacytoid-dendritic-cell→IFN- α innate axis producing storiform fibrosis and obliterative phlebitis), Type 2 (idiopathic duct-centric pancreatitis, IgG4-negative, neutrophil/granulocytic-epithelial-lesion-mediated, associated with inflammatory bowel disease), and Type 3 (immune-checkpoint-inhibitor-induced). It is polygenic/multifactorial (HLA-DRB1, FCGR2B, CTLA4, FCRL3), typically presents in older men with painless obstructive jaundice mimicking pancreatic cancer, is diagnosed via ICDC/HISORT criteria, and carries a favorable prognosis (~85% 10-year survival) despite ~40% relapse in Type 1, managed with corticosteroids, rituximab, and the newly FDA-approved anti-CD19 agent inebilizumab.